



Computer vision aided inspection on falling prevention measures for steeplejacks in an aerial environment

Qi Fang^{a,b}, Heng Li^{b,*}, Xiaochun Luo^b, Lieyun Ding^a, Hanbin Luo^a, Chengqian Li^a

^a School of Civil Engineering & Mechanics, Huazhong University of Science & Technology, Wuhan, China

^b Department of Building and Real Estate, The Hong Kong Polytechnic University, Hong Kong

ARTICLE INFO

Keywords:

Fall prevention
PPE
Automated monitoring
Computer vision
Deep learning

ABSTRACT

Falling from height accidents are a major cause of fatalities on construction sites. Despite a lot of research conducted on the enhancement of safety training and removal of hazardous areas, falling accidents remain a major threat for steeplejacks. According to NIOSH FACE reports, 75.1% of the fall from height decedents didn't use the Personal Fall Arrest Systems (PFAS), which shows insufficient supervision of the use of Personal Protective Equipment (PPE) by steeplejacks. Few scholars consider PFAS an important measure to prevent falls and the existing studies on PPE inspections showed that they were unsuitable for the scenarios faced by steeplejacks. This paper proposes an automated inspection method to check PPEs' usage by steeplejacks who are ready for aerial work beside exterior walls. An aerial operation scenario understanding method is proposed, which makes the inspection a preventative control measure and highly robust to noise. A deep-learning based occlusion mitigation method for PPE checking is introduced. We tested the performance of our method under various conditions and the experimental results demonstrate the reliability and robustness of our method to inspect falling prevention measures for steeplejacks and can help facilitate safety supervision.

1. Introduction

Falls from height is one of the main causes of fatalities in the construction industry [1]. According to the data provided by the Bureau of Labor Statistics, out of 937 fatal construction accidents in the United States in 2015, 350 of them were caused by falls [2]. According to the HSE's (Health and Safety Executive) summary, in Great Britain, falls from height also remained the biggest cause of injury in the construction industry during 2016 [3,4]. Similarly, the National Institute for Occupational Safety and Health suggest that fall accidents account for around 40% of fatal accidents and 30% of accidents resulting in injuries in Japan, and 40% of fatal accidents and 17% of non-fatal injuries in Korea [5]. In addition, fall protection remains at the top of the safety violations lists of OSHA's (Occupational Safety and Health Administration) reports since 2016 [6], where inadequate fall protection is one of the major causes.

Most fatal falling from height accidents are in fact preventable [1]. Plenty of standards and regulations about prevention measures have been implemented in various countries and regions to ensure the safety of steeplejacks. The *Practical Guide to Working At Height*, published by the Hong Kong Housing Authority [7], points out that appropriate implementation of barriers, fences, guard-rails and working platforms,

as well as Personal Fall Arrest Systems (PFAS), are necessary measures to prevent falls. The *Duty to Have Fall Protection* [8] and *Fall Protection Systems Criteria and Practices* [9] standards of the United States specify how to construct safe platforms under various working environments at height, as well as a detailed description of how to use personal protective equipment (PPE) for the work. In Great Britain, *The Work at Height Regulations* stipulate that 'every employer is required to carefully inspect the fall protection measures of the whole aerial workplace and steeplejacks shall select suitable PPE according to their trades' [10]. In summary, necessary inspections and appropriate use of PPE assists in the development of a safe working environment at height.

In view of the high frequency and severe consequence of falling from height accidents, a lot of research has focused on searching for every possible way to avoid such accidents from happening, e.g. by removal of hazardous areas, use of guardrail or safety net systems, administrative controls or fall portents detection. However, due to various unexpected situations, steeplejacks are inevitably faced with the risk of falling so the use of the PFAS is the protection measure most likely to save their lives. OSHA pointed out that PFAS are only required to be used in the working environment at height of 6 ft or more [11]. In fact, in NIOSH FACE reports [12], it was found that at the time of the fall, 54.2% of the fall decedents did not have access to PFAS; 23.1% of

* Corresponding author.

E-mail address: bshengli@polyu.edu.hk (H. Li).

the PFAS status is present but not in use; and only 2.2% of the decedents fall from height of < 6 ft. In other words, a total of 75.1% of the fall decedents are required to use PFAS but fail to do so.

Meanwhile, in certain specific situations, especially when decorating exterior walls, workers are occasionally exposed to dangerous working platforms (e.g. edges without barriers, fences, guard-rails etc.). Therefore, a new automatic inspection method is urgently needed to protect steeplejacks by enhancing the supervision of PFAS use at height. However, two aspects challenge the application of the automatic inspection of a PFAS by computer vision methods. First, the usage of PFAS is conditional and is only used when workers are at a height of 6 ft or more [11]. The automatic supervision method is required to determine whether a worker is engaged in works at height before inspecting their usage of PPE. On the other hand, the supervision of the PFAS is not limited to harness wearing but also rope anchor checking and wearing hardhats [11,13].

The aim of this paper is to promote the use of the PFAS in working environments at height beside exterior walls, since falling accidents remain the leading cause for work fatalities in the construction industry in many countries, including the U.K. (44%) and the U.S. (35%) [14–16]. Supervision enhancement of the use of the PFAS in aerial work is one of the most effective ways to ensure the safety of steeplejacks. Few studies considered taking approaches to strengthen the inspection of the PPE use by steeplejacks and the existing studies on PPE monitoring are not yet applied to PFASs. Accordingly, this paper proposes a novel method, based on computer vision, to automatically inspect the appropriate usage of PFASs among steeplejacks before they enter an aerial working environment. The core of this paper lies in two parts as follows: First, considering the scope of usage of PPE, this paper proposes a method of scenario recognition for aerial working environments, which is used as the basis for judging whether PFASs are needed. Second, we have a trained deep learning model to identify multiple PPEs. And the inspection of PPEs includes not only harness checking but also webbing, anchoring and wearing hardhats. Our method can identify and distinguish multiple combinations of unsafe behavior.

2. Literature review

This study focuses on preventing accidents caused by falls from heights for steeplejacks through inspecting the appropriate use of PPEs. Thus, firstly, we reviewed solutions provided by related literature for the falls from height prevention in Section 2.1. Then, related research on PPEs' inspection is discussed in Section 2.2. In fact, few scholars considered the PFAS as an important measure to prevent fall accidents and the existing studies on PPEs' inspection did not meet the requirement of specific scenarios for steeplejacks. Therefore, an introduction to the development of computer vision based object detection methods, which were used in our solution, is followed in Section 2.3.

2.1. Related research of fall prevention

According to the provisions of OSHA [17] and the summary of Esmaeil, Hallowell, et al. [18], removal of hazardous areas, use of guardrails or safety net systems, administrative controls and PFAS, constitute the primary protection for steeplejacks.

Scholars have done a lot of research on how to prevent construction workers from falling at height. On the one hand, removal of hazardous areas by design optimization is a feasible solution. Qi, Issa, et al. [19] expand on the IFC hierarchy to conduct compliance checking and optimize building designs for safety, which provides an opportunity to prevent workers from falling. Zhang, Sulankivi, et al. [20] developed a BIM based prototype with safety rule checking algorithms which can identify and eliminate potential fall hazards in the planning phase. Wang, Pradhananga, et al. [21] proposed a laser scanning method to identify fall risks by analyzing geometrical properties during the construction phase. On the other hand, some of the scholars have

committed to the appropriate use of guardrail or safety net systems to protect workers. Navon and Kolton attached sensors to guardrails to inspect their installation, and warnings will be issued whenever guardrails are missing or different from the planned ones [22,23]. Zuluaga and Albert [24] use virtual prototyping methods to check bridge guardrails' usage. Cheung and Chan [25] directly invented a Rapid Demountable Platform (RDP) device, which can be flexibly applied to prevent external workers from falling from height. Furthermore, administrative controls are another method to help reduce falling accidents by improving workers' safety awareness. Kaskutas, Dale, et al. [26] found that groups trained by a foreman and apprentice in fall prevention, greatly improved the safety awareness and working environment of workers. Lin, Migliaccio, et al. [27] used a 3D training platform to promote the knowledge of canonical working procedures of falling protection. Evanoff, Dale, et al. [28] focus on improving the workers' participation in training, such as hands-on practice, simulations and reality-based training. These improved training methods aimed at enhancing the safety behavior of workers when working at height. In addition, many researchers recognized workers' postures and movement patterns by making them wear devices and then analyzing the fall portents, thus reducing fall accidents [29–33].

Despite PFAS being one of the most important measures in preventing fall accidents, few scholars work on the appropriate usage checking of PFAS for the steeplejacks.

2.2. Related research on PPEs inspection

Despite few scholars focusing on the inspection of PFAS usage for steeplejacks, a lot of research proceeds in other PPEs' inspection for workers on construction sites. Here, PPEs refer to garments or equipment designed to protect the workers from being injured, mainly including fall arrest systems, protective clothing, helmets, goggles and so on [34]. The state-of-the-art studies for PPEs' inspections are introduced below.

Sensor based methods are widely used in PPEs' inspections due to their portability and flexibility. Barro-Torres, Fernández-Caramés, et al. [35] attached RFID tags to all PPEs and a RFID reader was given to every worker at the same time. This way, the readers attached to the workers could detect the presence of PPE that was worn by the workers. Kelm, Laußat, et al. [36] used automated identification (ID) and information technologies (IT) to design a RFID portal that was positioned at the entrance of the construction site to check PPE compliance of personnel. If a RFID portal is placed beside the window, we can check the presence of PFAS when a steeplejack is going out through the window. However, whether the harness is worn by the steeplejack and the hook has been anchored or not, can hardly be distinguished. Podgórski, Majchrzycka, et al. [37] use the Internet of Things (IoT) technologies to create smart working environments in which not only PPEs' information, but also hazardous and strenuous factors, such as noise, exposure to toxic chemical substances, optical radiation and high or low temperatures are monitored. Dong, He, et al. [38] combine pressure sensing and Bluetooth to assess how the hardhats are worn. The sensor-based methods listed above are limited by their intrusive nature, which is highly dependent on workers' active cooperation. Moreover, it's hard to confirm the proper use of PPEs.

Besides sensor-based methods, the application of computer vision techniques in PPE monitoring is becoming more and more popular. Some scholars use vision based methods to determine whether workers are wearing hardhats or not. Du, Shehata, et al. [39] determine whether the PPEs have been worn by comparing the color features of the target area on workers with the predefined template of the PPEs. Shrestha, Shrestha, et al. [40] improved the above method by adding more feature information besides color to improve the recognition accuracy of PPEs. These handcrafted features defined by researchers are based on an assumption that the template features are the same as the practical objects, which is often unrealistic. Park, Elsafty, et al. [41] proposed a

system architecture of PPE use detection in which Histogram of Oriented Gradient (HOG) features are drawn from the video images and input into the Support Vector Machine (SVM) for classification. Despite the accuracy that has exceeded Du and Shrestha's research, HOG methods are still limited by low accuracy (10.2% precision in the PASCAL VOC 2006 [42]), which makes it unsuitable for practical use on site. Along with the rapid development of deep-learning techniques, Fang, Li, et al. [43] put forward a deep learning based method to detect non-PPE use on site. The proposed method resulted in high precision, high recall and fast speeds among various kinds of visual conditions on construction sites. However, all the methods above were only designed to detect target PPEs without considering their usage scopes and proper manner of use. For example, hardhats are required to be worn everywhere on site, while PFAS are only required when workers are at the height of 6 ft or more [11]. On the other hand, the proper use of PPEs should also be inspected.

In summary, the existing studies on PPEs' inspections show that they aren't suitable for steeplejacks, since they neither recognize PPEs' usage scenarios nor distinguish the proper usage of PPEs. PPEs are the last safeguard for steeplejacks. Considering few scholars emphasized its importance in preventing fall accidents and the existing studies on the PPEs' inspection program didn't meet the specific requirements of steeplejacks, there is value in proposing an advanced method to automatically monitor the PPEs use before steeplejacks enter aerial working areas. We have considered deep-learning based techniques as a feasible solution because of low extra costs (construction sites are generally equipped with surveillance cameras), simple structures and high performance in vision analysis. Meanwhile, the field of computer vision has developed rapidly [44,45] and the application of vision-based methods to monitor safety is expected to achieve great advances in the near future [46–49]. The vision based approach enables us to create an effective monitoring environment for steeplejacks to ensure the proper use of PPEs. Next, we'll review the development of computer vision algorithms that we have employed in this study.

2.3. Related deep learning algorithms in computer vision

Inspection of the appropriate usage of PFAS requires several techniques in computer vision: object classification, object detection and tracking algorithms are introduced in the following paragraph.

CNN (convolutional neural network) is mainly used for image classification [50], which aims to determine the category of an object in a close-up image. Meanwhile, CNN is the basic kernel of deep learning methods used in computer vision. A complete CNN consists of multiple convolutional layers, rectified linear units (ReLU), pooling layers as well as a fully connected layer. The convolutional layers are the core building block of a CNN. The parameters of learnable filters in these layers are fine-tuned during the training phase. As a result, the learned filters are used to detect and extract specific types of features by convolutional operation with the input images. The ReLU layer enhances or suppresses the signals from convolutional layers by nonlinear mapping. The pooling layer can progressively reduce the spatial size and the number of parameters, which is helpful to avoid over fitting. The high-level reasoning is done by the fully connected layer. It outputs the category confidence for each object of the input image.

As CNN processes close-up images of only one specific object, additional algorithms are needed to extract them. Therefore, object detection methods separate objects from the background first and then determine their categories through CNN. Girshick, Donahue, et al. [51], proposed a R-CNN model in which the localization and segmentation process is added to the framework of basic CNN. He, Zhang, et al. [52] added a SPP (spatial pyramid pooling) layer in the R-CNN model to avoid graphics distortion that is caused by warping or cropping. Ren, He, et al. [53] presented Fast R-CNN that accelerated the processing speed of RCNN by reconstructing the internal structure in the model. Ren, He, et al. [53] put forward the Faster R-CNN method, which

extracted region proposals through CNN and improved the detection precision. Compared to Faster R-CNN, YOLO [54] is a faster but lower precision algorithm with problems such as missing the detection of small objects. Finally, SSD [55] inherits the advantages of both Faster R-CNN and YOLO, and achieves both high precision and fast speed.

Therefore, we have considered SSD as the object detection solution that will be employed in this paper. In a SSD framework, similar to other object detection methods, a feature map is extracted through a VGG16 network. Next, SSD utilize feature maps from several different layers to jointly analyze whether a default box is an object or not. Since feature maps from different levels are considered to have different receptive field sizes [56] and the previous object detection methods didn't utilize such advantages, the feature maps' capture mode of SSD is superior and analyzes an image more comprehensively. Meanwhile, default boxes of different layers represent different pixel sizes. Therefore, SSD can detect objects of different scales and won't ignore tiny objects as with the YOLO method. Multiple convolutional feature layers are positioned at the end of the network to synthesize feature maps from different layers. Compared with using a fully connected layer, adding multiple convolutional feature layers to the end can greatly reduce the number of parameters and improve speed.

Multiple object tracking (MOT) is another frequently used technique in the computer vision field [57]. The same objects across different frames are associated with MOT methods and coherent information on the dynamic changes in a video sequence is extracted at the same time. There are two alternative types of candidate MOT methods in this study. Correlation filter based tracking methods [58–61] achieve higher accuracy at the expense of speed, which limits their use in real-time application. In comparison, simple online and real-time tracking (SORT) [57] is a much simpler framework that achieves a favorable performance. Due to its real-time processing speed and low training cost, SORT is employed in this study.

SORT tracker predicts a new position of the moving object by combining the location information in the current frame and the displacement direction and speed from previous frames. The following is the framework of SORT: Kalman filter [62] is used in the prediction algorithm, then the assignment of the objects across the frames is solved by a Hungarian algorithm [63] by linking the new location of an unknown object detected in an adjacent frame and the predicted location from the Kalman filter. Therefore, all the objects are managed and tracked in the video sequence.

3. Methodology

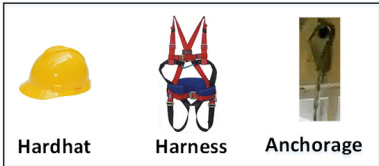
This paper presents a novel framework to inspect the appropriate usage of PPEs by steeplejacks working at height beside exterior walls. Fig. 1. illustrates the overall framework of the proposed methods involving these two steps.

3.1. Understanding aerial operation scenarios

There are two kinds of camera layout schemes to capture the videos of steeplejacks, which will be discussed here in detail. The first scheme is to arrange cameras outside the building and capture videos from a far-field perspective. Since the dense safety net and scaffolding severely obstruct the vision of cameras and the long-distance greatly impacts the resolution, it is hard for outdoor cameras to collect high quality videos. Insufficient resolution and severe obstruction of videos impacts the performance of computer vision-based technologies. Considering the obstructed vision of the cameras, the worker may have been engaged in an aerial working area for a long time before he was found not to be using any PPE. A better scheme would warn the steeplejacks before they enter the danger zone. Therefore, this scheme isn't applicable to construction sites mainly because of the obstruction and low resolution.

The second scheme is to arrange the cameras inside the buildings and facing the windows. Steeplejacks always go through a window to

Step 2:
Inspect the
appropriate
usage of PPEs



Step 1:
Ensure the
worker is
entering an
aerial operation
scenario

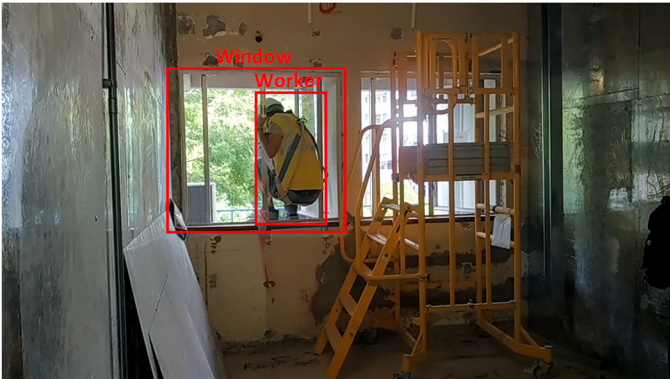


Fig. 1. Overall framework of the proposed method.



Fig. 2. The difference between ground truths and detected bounding boxes.

reach the outside of a building for aerial work if the main structure of the building has been completed. This is a common procedure for the steeplejacks who are engaged in equipment installation and exterior wall decoration during either the construction stage or the renovation of an old building. Therefore, if we install cameras in rooms that face the windows, every worker who is trying to get to the outside of a building through a window will be captured by the surveillance cameras. The distance between the camera and steeplejacks indoors is much less than the distance between the camera and steeplejacks outdoors, and there are hardly any obstructions in the room, such as dense safety nets or scaffolding, which makes it much easier for indoor cameras to obtain high quality videos. Meanwhile, inspecting the usage of PPEs by steeplejacks before they reach a danger area is a preventative control that guarantees the safety of steeplejacks. Therefore, our method is designed to inspect the steeplejacks on the condition that it is the scenario of a worker who's going out through a window.

In comparison, scene recognition, which requires contextual reasoning in addition to object detection, is still an open challenge in recent research for computer vision techniques [64]. Here, we propose an aerial operation scenario understanding method by combining the computer vision technologies and an aerial scenario calibration (ASC)

Table 1

Six categories of positions between a steeplejack and a window.

No.	The steeplejack gets close enough to one edge of the window				Demonstration
	Top	Left	Right	Bottom	
1	x	x	x	✓	
2	✓	x	x	✓	
3	x	✓	x	✓	
	x	x	✓	✓	
4	✓	✓	x	✓	
	✓	x	✓	✓	
5	x	✓	✓	✓	
6	✓	✓	✓	✓	

classifier to recognize steeplejacks and inspect whether they have carried out necessary safety measures before they take part in tasks at height.

As the basis of the next step, the same objects (windows and workers) are required to be detected and matched in different frames. First, an object detection method (SSD [55]) is used to detect key objects in an image frame of surveillance videos. Then, a SORT [57] based tracking method is used to associate the same objects across frames in a video sequence.

Spatial interaction between objects is then analyzed as the key

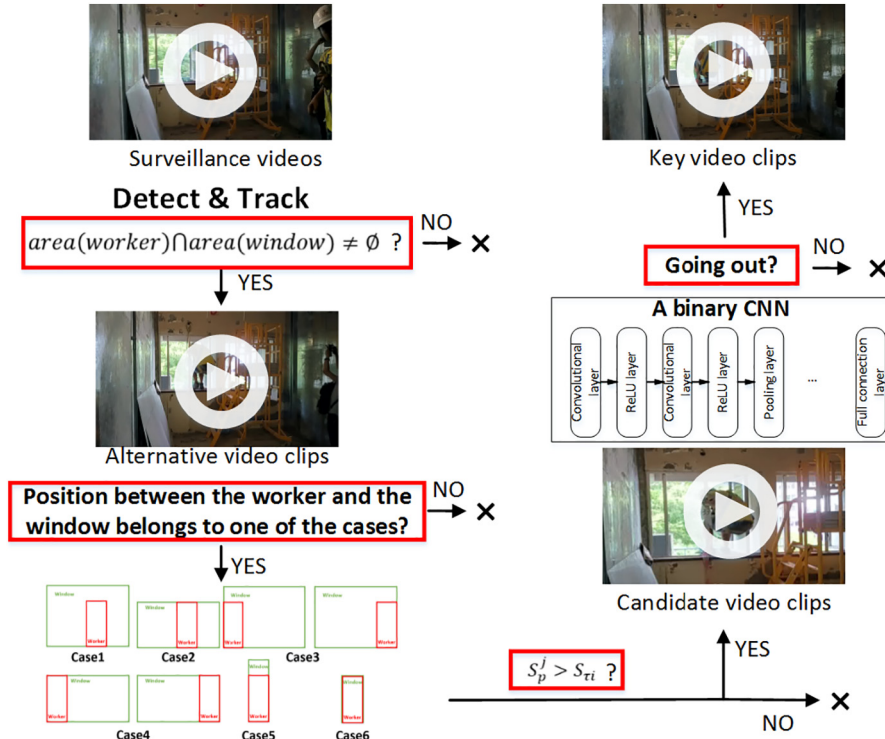


Fig. 3. The procedure of aerial operation scenarios understanding.

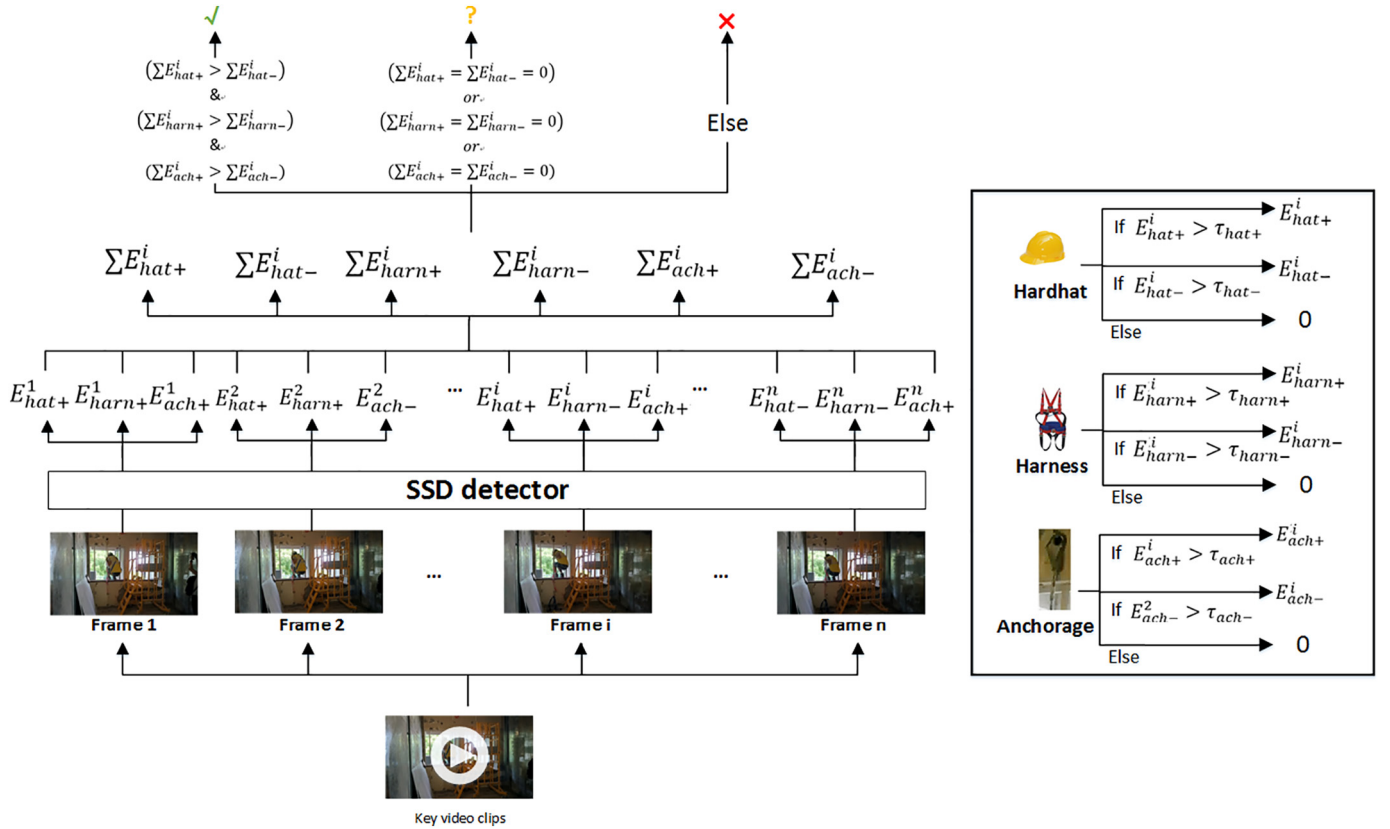


Fig. 4. The module of PPEs' inspection for steepplejacks.

context information for scenario understanding. For a worker p and a window q , if the spatial interactions between them in consecutive frames all meet the condition that $area(p) \cap area(q) \neq \emptyset$, then the consecutive frames will be extracted as an alternative video clip for the

worker. The alternative video clip of the worker p is annotated as $\Gamma = \{f_1, f_2, \dots, f_i\}$. In each frame, the spatial interaction between the worker and the window is calculated as Formula 1. Here, we define spatial interaction as S_{pq} , which indicates the interaction between the

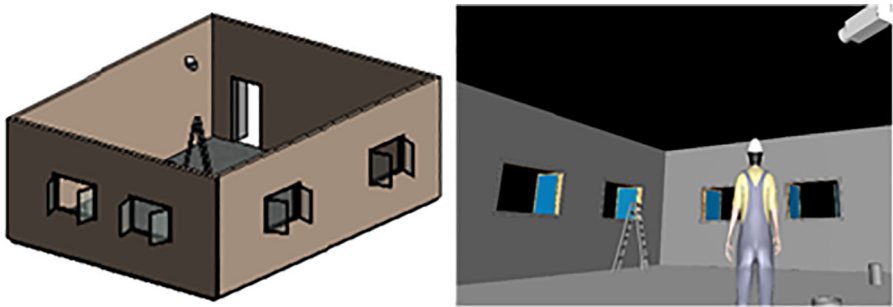


Fig. 5. Example of a camera's placement in a room.

Table 2
Information of training dataset.

Training model	Objective	Training dataset	Number of images
ASC classifier	Aerial operation scenario recognition	A worker is totally inside a window	9000 (about 1500 for each category listed in Table 1)
		A worker intersects with a window but isn't inside it	9000 (about 1500 for each category listed in Table 1)
Binary CNN model	Elimination of distractors where steeplejacks getting in through a window	A steeplejack is getting in through the window	2000
		A steeplejack is getting out through the window	2000
SSD model	Aerial operation scenario recognition	Worker	2000
	PPEs inspection	Window	2000
		A worker wearing a hardhat	2000
		A worker not wearing a hardhat	2000
		A worker wearing a harness	2000
		A worker not wearing a harness	2000
		Anchorage linked with a webbing	2000
	Anchorage not linked with a webbing	2000	

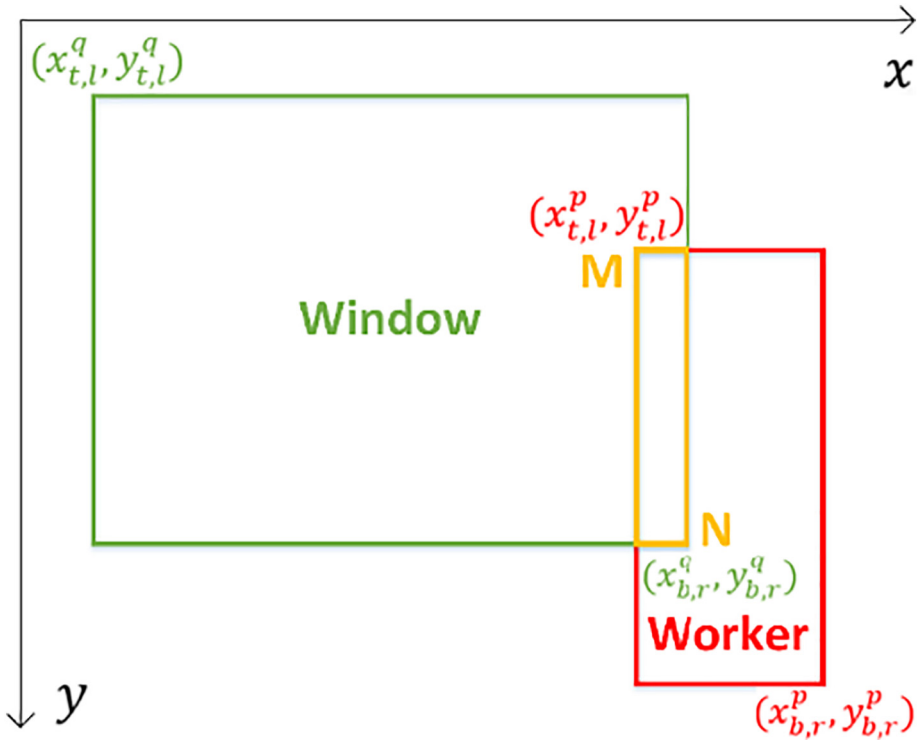


Fig. 6. Spatial interaction of two bounding boxes.

Table 3
Spatial interaction computation procedure.

No.	Worker				Window				M		N		Si
	$X_{t,l}$	$Y_{t,l}$	$X_{b,r}$	$Y_{b,r}$	$X_{t,l}$	$Y_{t,l}$	$X_{b,r}$	$Y_{b,r}$	XM	YM	XN	YN	
1	15	280	384	1278	140	298	529	657	140	298	384	657	0.2379
2	35	212	393	1257	148	305	533	661	148	305	393	661	0.2331
3	35	240	376	1267	155	303	541	660	155	303	376	660	0.2253
4	36	267	395	1217	144	310	530	667	144	310	395	667	0.2627
5	40	236	385	1205	156	301	543	658	156	301	385	658	0.2445
6	32	217	385	1248	152	306	540	665	152	306	385	665	0.2298
7	36	190	383	1246	142	297	530	657	142	297	383	657	0.2368
8	8	247	393	1278	164	303	546	656	164	303	393	656	0.2037
9	36	223	376	1251	146	302	524	651	146	302	376	651	0.2297
10	109	234	276	519	148	301	526	650	148	301	276	519	0.5863
...	...	...	...	...	...	...	...	...	...	...	...	...	...

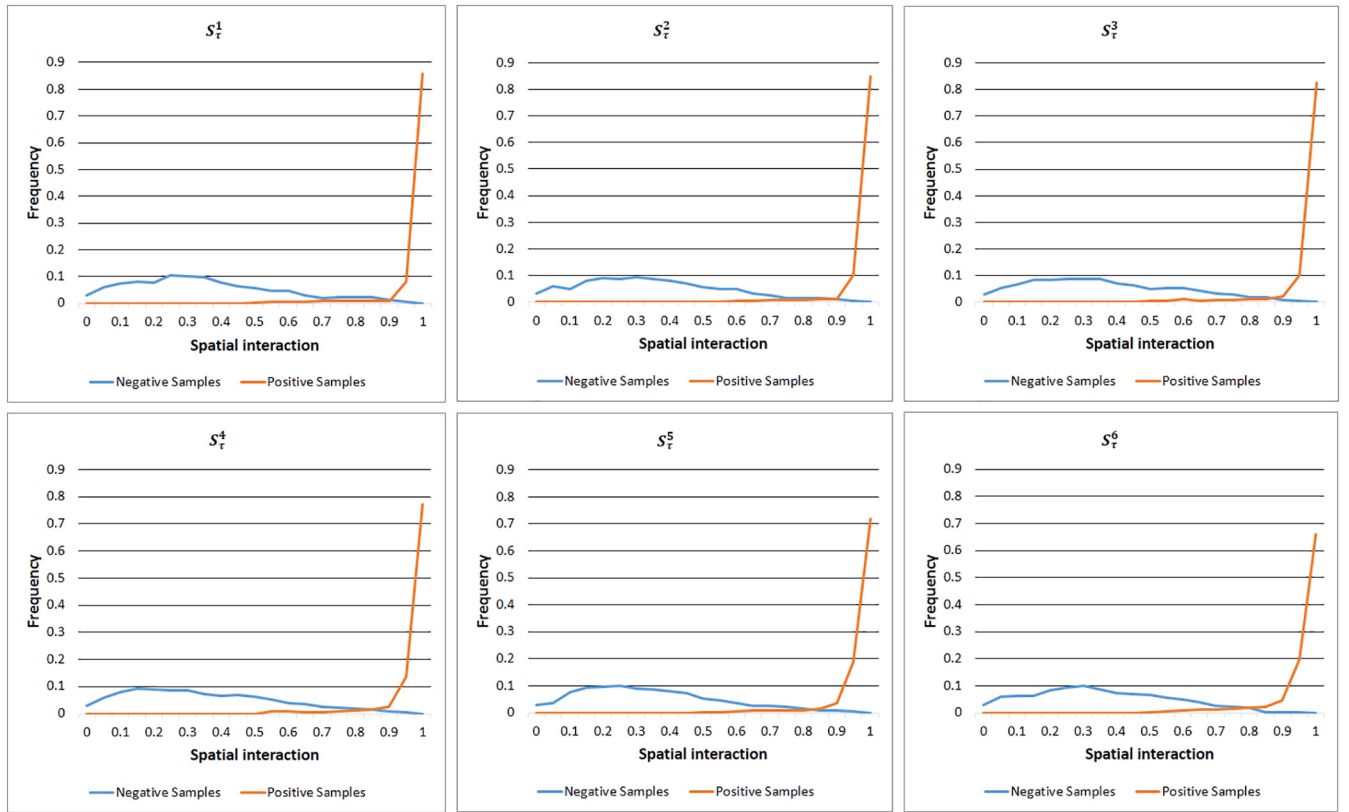


Fig. 7. Probability density distribution of positive/negative samples.

Table 4
Information of testing dataset.

Testing objective	Classifications	Scenarios in each category	Number of testing video clips
Impact of illumination	4 different luminous flux	24	1116
Impact of worker's characteristics	4 different workers	24	1116
Impact of window's characteristics	6 different windows	16	1116
Impact of occlusion	Different occlusion ratios	/	400

worker p and the window q .

$$S_{pq} = \frac{\text{area}(p) \cap \text{area}(q)}{\text{area}(p)}$$

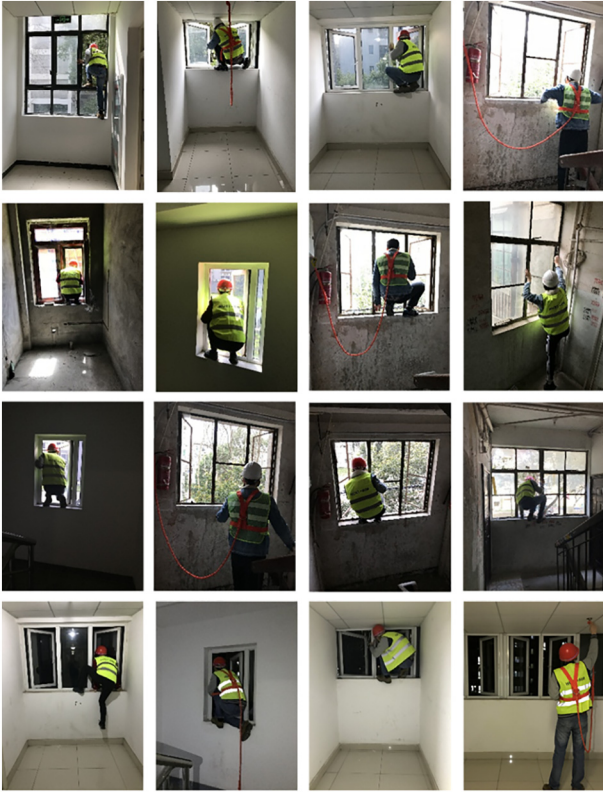
Where $\text{area}(\ast)$ represents the area of the bounding box of a worker or an object. Note that we define box areas by units of pixel^2 .

Ideally, detecting a frame that the worker is totally inside of is sufficient proof that he is trying to go out through the window. Therefore, given a video clip of a worker in ideal status, if the max value of S_{pq} equals 1, there must be a worker going out through the window. However, inaccuracies in training and limitations of the model will produce noise and errors [65], which means differences exist in both scale and spatial location between the detected bounding boxes from the computer vision and the actual outlines of objects from human eyes. Fig. 2 represents the difference between ground truth and detected bounding boxes of both windows and workers (the green boxes indicate ground truth and the red boxes indicate detected bounding boxes).

Table 5

Test results under different illumination, worker's and window's characteristics.

No.	Illumination	Worker	Window	Number of video clips	Number of inappropriate-PPEs-use steeplejacks	TP	FP	FN	Precision	Recall rate
1	Daylight: 750–1500 (lux)	P1	W1	9	7	7	1	0	0.875	1.000
2	Daylight: 750–1500 (lux)	P1	W2	12	10	9	0	1	1.000	0.900
3	Daylight: 750–1500 (lux)	P1	W3	12	11	11	1	0	0.917	1.000
4	Daylight: 750–1500 (lux)	P1	W4	15	14	13	1	1	0.929	0.929
...	...	...	...	...	...	...	...	...	...	...
31	Daylight: 200–750 (lux)	P2	W1	13	12	12	0	0	1.000	1.000
32	Daylight: 200–750 (lux)	P2	W2	15	14	14	0	0	1.000	1.000
33	Daylight: 200–750 (lux)	P2	W3	9	7	6	0	1	1.000	0.857
34	Daylight: 200–750 (lux)	P2	W4	11	10	10	0	0	1.000	1.000
...	...	...	...	...	...	...	...	...	...	...
63	Daylight: 20–200 (lux)	P3	W3	10	8	7	0	1	1.000	0.875
64	Daylight: 20–200 (lux)	P3	W4	11	9	7	1	2	0.875	0.778
65	Daylight: 20–200 (lux)	P3	W5	10	8	7	1	1	0.875	0.875
66	Daylight: 20–200 (lux)	P3	W6	11	10	8	1	2	0.889	0.800
...	...	...	...	...	...	...	...	...	...	...
93	Lamplight: 50–500 (lux)	P4	W3	8	6	5	0	1	1.000	0.833
94	Lamplight: 50–500 (lux)	P4	W4	8	6	6	1	0	0.857	1.000
95	Lamplight: 50–500 (lux)	P4	W5	15	13	12	1	1	0.923	0.923
96	Lamplight: 50–500 (lux)	P4	W6	10	9	9	1	0	0.900	1.000

**Fig. 8.** Image frame examples under 4 different illumination conditions.

These errors in scale and location may cause a worker's bounding box to exceed the window's area. Thus, from the view of computer vision, the case that the max value of $S_{pq} < 1$ when a worker is actually inside a window is common. It's inaccurate to judge the scenarios of aerial work only by $S_{pq} = 1$ and a new algorithm, considering the error influence generated in computer vision detection process is needed.

Thus, an ASC classifier is employed to solve the above problem. Given a video clip in which an intersection exists between a worker and a window, an ASC classifier can determine whether the worker is inside the window according to the knowledge learnt from abundant training samples. The advantage of the ASC classifier is that it's highly robust to noise and errors generated in vision based detection and tracking algorithms, which makes it able to classify different scenarios precisely.

Here, S_p^i indicates spatial interaction between the detected bounding boxes of a worker p and a window q in the i^{th} frame of an alternative video clip. H_T represents the hypothesis that the worker is going out through the window, and H_F represents the hypothesis that the worker is not going out through the window. The problem of judging whether the worker is going out through the window amounts to a naive Bayesian decision formulation [66].

$$P(H_T | S_p^i) = \frac{P(S_p^i | H_T) \times P(H_T)}{P(S_p^i)} \quad (2)$$

$$P(H_F | S_p^i) = \frac{P(S_p^i | H_F) \times P(H_F)}{P(S_p^i)} \quad (3)$$

Given a prior probability that $P(H_T) = P(H_F)$, we can get Formula 4:

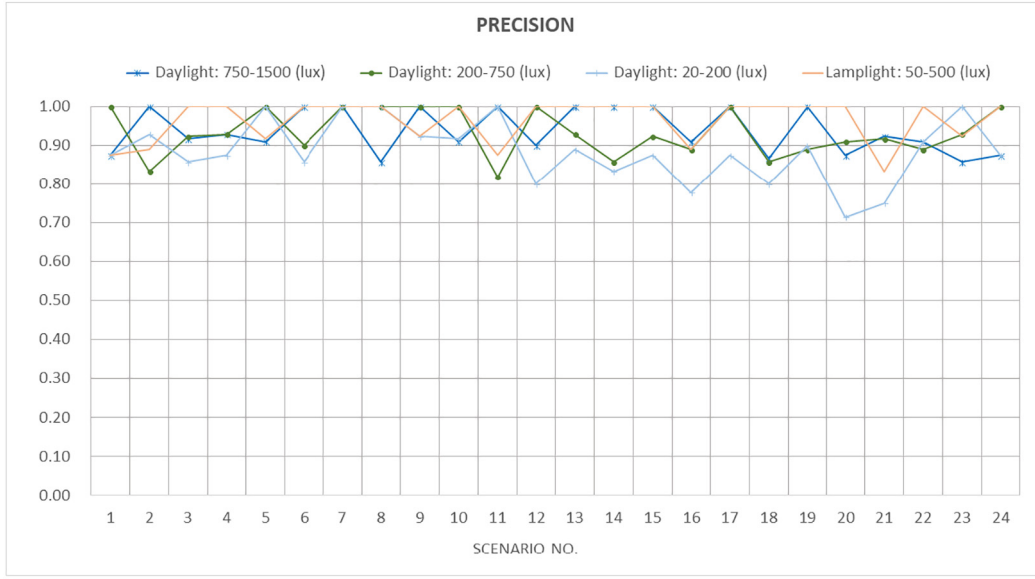
$$\frac{P(H_T | S_p^i)}{P(H_F | S_p^i)} = \frac{P(S_p^i | H_T)}{P(S_p^i | H_F)} \quad (4)$$

Here, $J(S_p^i)$ is used to indicate whether the worker p in the alternative video clip is going out through the window:

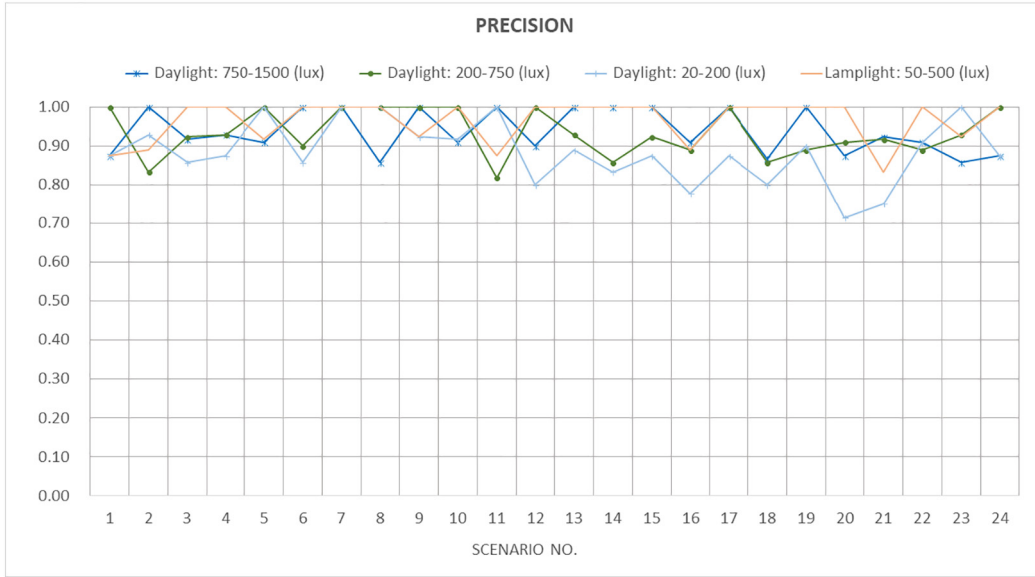
$$J(S_p^i) = \log \frac{P(S_p^i | H_T)}{P(S_p^i | H_F)} \quad (5)$$

To calculate $J(S_p^i)$, $P(S_p^i | H_T)$ and $P(S_p^i | H_F)$ need to be solved first. A large number of samples, $X = (x_1, x_2, \dots, x_n)$, are collected and categorized into H_T or H_F to observe their probability density distribution. Each sample is annotated with $x_j(S^j, C^j)$, where S^j is spatial interaction between the detected bounding boxes of a worker and a window in frame j , which is calculated by Formula 1, and C^j refers to the category of sample (H_T or H_F). From the statistics analysis of these training samples, we concluded the distribution functions for S in H_T and H_F hypothesis respectively: $F_T(S)$ and $F_F(S)$. Let $f_T(S)$ and $f_F(S)$ be the corresponding probability density of $F_T(S)$ and $F_F(S)$. Therefore, given $S = S_p^i$, $P(S_p^i | H_T)$ and $P(S_p^i | H_F)$ are calculated respectively by Formula 6 and 7.

$$\begin{aligned} p(S_p^i | H_T) &= \lim_{\Delta S \rightarrow 0} \frac{P(S_p^i \leq S \leq S_p^i + \Delta S | H_T)}{\Delta S} \\ &= \lim_{\Delta S \rightarrow 0} \frac{F_T(S_p^i + \Delta S) - F_T(S_p^i)}{\Delta S} = f_T(S_p^i) \end{aligned} \quad (6)$$



(a).



(b).

Fig. 9. (a). Precision ratios under different illumination levels
(b). Recall rate ratios under different illumination levels.

Table 6
Precision and recall rate ratios under different illumination levels.

No.	Illumination	Average precision	Average recall rate
1	Daylight: 750–1500 (lux)	0.938	0.956
2	Daylight: 200–750 (lux)	0.933	0.953
3	Daylight: 20–200 (lux)	0.882	0.885
4	Lamplight: 50–500 (lux)	0.931	0.963

$$\begin{aligned}
 p(S_p^i | H_F) &= \lim_{\Delta S \rightarrow 0} \frac{p(S_p^i \leq S \leq S_p^i + \Delta S | H_F)}{\Delta S} \\
 &= \lim_{\Delta S \rightarrow 0} \frac{F_F(S_p^i + \Delta S) - F_F(S_p^i)}{\Delta S} = f_F(S_p^i)
 \end{aligned} \quad (7)$$

Therefore, given the probability density distribution curves of $f_T(S)$ and $f_F(S)$, $J(S_p^i)$ is obtained by Formula 8.

$$J(S_p^i) = \log \frac{f_T(S_p^i)}{f_F(S_p^i)} \quad (8)$$

Given S_r where it meets the requirement that $J(S_r) = 0$. It is noted that location errors and scaling errors are the main reasons that cause $S_r < 1$. Only if the edge of a steeplejack's bounding box gets close enough to a window's (for example, $|x_b, r^q - x_b, r^p| < \omega(x_b, r^p - x_t, l^p)$, where ω is a predefined parameter, and other variables can be referred to Fig. 6), the above errors might lead to a false recognition of the steeplejack's edge exceeding the window's bounding box. Therefore, in the circumstances where the steeplejack is actually inside the window, different positions between the steeplejack and the window might cause different S_r . As shown in Table 1, six categories of positions are defined in this paper, and each class is trained separately with its own dataset. Finally, we get six different S_r corresponding to the different positions.

The entire program of detecting a steeplejack can be concluded as



Fig. 10. Image frame examples including different physical characteristics of workers.

follows: First, all frames meet the condition that $\text{area}(p) \cap \text{area}(q) \neq \emptyset$ will be extracted as an alternative video clip. Given an arbitrary frame of a video clip, if the position of a worker and a window satisfies $S_p^i > S_r^j$ (noted that different positions correspond to different S_r^j), this frame is selected as a candidate frame. Then the appearance features of the worker in this frame is extracted by a simple binary CNN model to further determine whether he is getting out or in through the window. Only the frames in which the scenario is determined as getting out will be preserved and further examined in the next section, while the others will be eliminated. All the frames to be inspected are defined as key frames, which means a steeplejack getting out through a window is detected in all these frames. The complete process of understanding aerial operation scenarios is presented in Fig. 3.

3.2. Inspection of appropriate usage of PPEs

The previous section provides an analysis of scenarios to determine if a worker is trying to get through a window to engage in aerial work outside, and this section applies a SSD detector to inspect the appropriate usage of both FPAS and hardhats. Appropriate usage of FPAS requires a steeplejack to wear the harness and hang the connector with a webbing on to the anchorage fixed on the wall. As for hardhats, steeplejacks are asked to wear them on their head rather than hold them in their hands or lay them down on the ground.

The specific process of inspecting the appropriate usage of PPEs is represented in Fig. 4. First, we trained a SSD model to detect six different scenarios as follows: a worker with hardhat, a worker without hardhat, a worker with harness, a worker without harness, an anchorage that is linked with the connector by a webbing and an anchorage that isn't. However, due to the influence of poor illumination and occlusions, the target objects (hardhat, harness and webbing) may not be detected simultaneously in an image frame for a steeplejack who has been equipped with PPEs appropriately. Also, the result in one frame

may not always be accurate. Therefore, the PPEs' inspection program will be conducted for all key frames in the previous section. As mentioned before, a SORT based tracking method can associate the same steeplejack in a video sequence.

Take hardhat-wearing inspection as an example. Let E_{hat+}^i (E_{hat+}^i should be above the confidence threshold τ_{hat+}) be the confidence of detecting a steeplejack wearing a hardhat in frame i , and E_{hat-}^i (E_{hat-}^i should be above the confidence threshold τ_{hat-}) be the confidence of detecting a steeplejack not wearing a hardhat in frame i . There is one more situation where neither of these two cases is detected in frame i due to occlusion, and no result is counted in such a situation. Then, we sum up E_{hat+}^i and E_{hat-}^i in all key frames respectively. If $\sum E_{hat+}^i > \sum E_{hat-}^i$, the steeplejack is considered to have been wearing a hardhat, and vice versa. The inspection process on harness-wearing and anchorage use are the same as for hardhat-wearing inspection. Only if $(\sum E_{hat+}^i > \sum E_{hat-}^i) \& (\sum E_{harn+}^i > \sum E_{harn-}^i) \& (\sum E_{ach+}^i > \sum E_{ach-}^i)$ are all true in a video clip, the steeplejack is allowed to continue to enter the aerial environment. If $(\sum E_{hat+}^i = \sum E_{hat-}^i = 0)$ or $(\sum E_{harn+}^i = \sum E_{harn-}^i = 0) \text{ or } (\sum E_{ach+}^i = \sum E_{ach-}^i = 0)$ is satisfied in a video clip, it indicates that a severe occlusions problem exists in the video clip, which causes difficulties in detection for the computer vision technique. Otherwise the steeplejack will be alerted by a loudspeaker placed beside the window.

4. Experiments and results

4.1. Experimental preparation

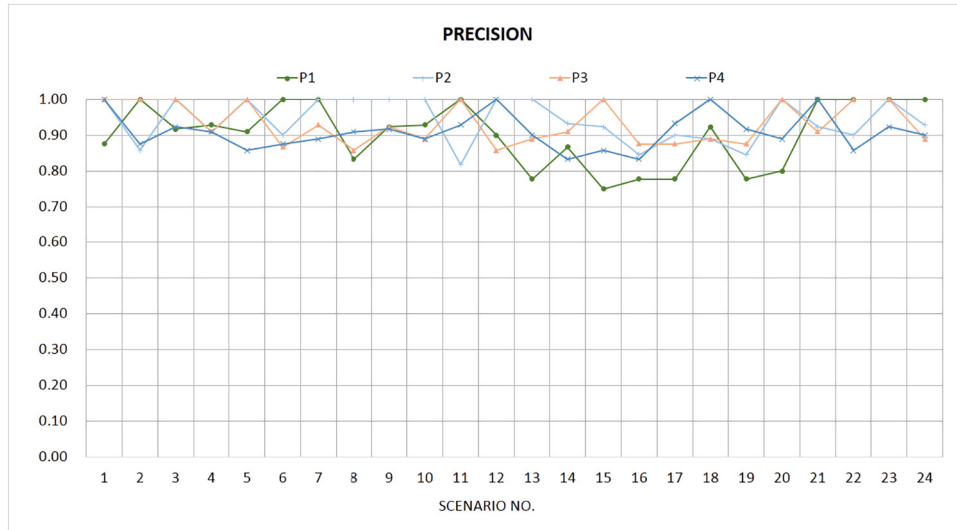
4.1.1. Placement scheme of cameras

As discussed in the methodology section, we used real-time surveillance videos, captured by indoor cameras, to inspect fall protection measures for steeplejacks. To achieve a better performance, the placement scheme of cameras is required to follow four principles: 1. monitor every steeplejack, 2. ensure sufficient resolution, 3. minimize occlusion, and 4. reduce the cost of equipment. Therefore, we installed one camera for each room at the height of about two meters to minimize occlusion, as shown in Fig. 5. The specific location of the camera was varied to coincide with the location of windows. If the room is so large that it causes insufficient resolution, or there are so many windows that a camera can't cover all of them, increasing the number of cameras is the recommended practice to apply to specific circumstances.

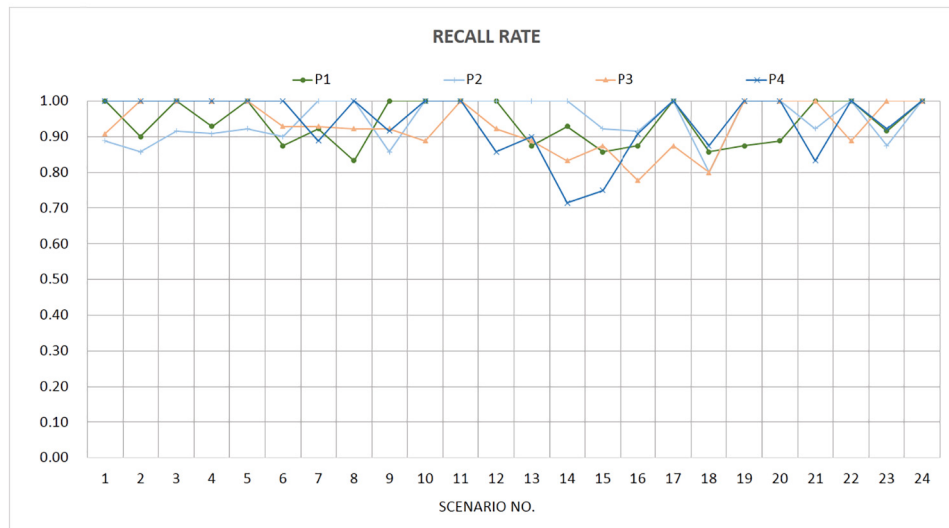
4.1.2. Training process

The training dataset consists of three parts corresponding to three models to be trained separately, see Table 2. First, image frames were collected in which workers intersect with windows so as to extract spatial interaction between the worker and the window to train the ASC classifier. Second, all of the positive samples ($S_p^i > S_r^j$) were divided into two classes (getting out or getting in) to train a binary CNN model. Third, in the aerial scenarios recognition program, two kinds of images (workers and windows) are collected in the dataset. In the PPE inspection program, six classes of different images (cropped images of workers with/without hardhat and harness, anchorages linked/not linked with webbing) are added to train a SSD model.

Here, we will introduce the detailed process of training the ASC classifier. The ASC classifier learns knowledge of classification by distinguishing the features of positive samples from negative samples in the training dataset. Therefore, the collection of a thorough training dataset with diversified samples is a key step for a well-trained ASC classifier. First, we divided image frames, in which bounding boxes of workers and windows intersect into the six categories mentioned in Table 1 ($\omega = 0.1$), and each category is trained separately by calculating the distribution of S_p^i using Excel and Origin [67]. Then image frames in the same category were manually annotated as two classes. We defined the classification principle as follows: those frames where a



(a).



(b).

Fig. 11. (a). Precision ratios under different physical characteristics of workers
(b). Recall rate ratios under different physical characteristics of workers.

Table 7

Precision and recall rate ratios under different physical characteristics of workers.

No.	Worker	Average precision	Average recall rate
1	P1	0.903	0.939
2	P2	0.941	0.945
3	P3	0.931	0.932
4	P4	0.909	0.940

worker is totally inside the window were classified into positive samples, while the others were classified into negative samples.

Next, we calculated the spatial interaction for each sample according to Formula 1, since the SSD detector returns the top left coordinates $(x_{t,l}, y_{t,l})$ and the bottom right coordinates $(x_{b,r}, y_{b,r})$ of the bounding box, we calculated the $area(p) \cap area(q)$ according to the following steps:

1. Calculate the coordinates of M and N, and $x_M = \max(x_{t,l}^p, x_{t,l}^q)$, $y_M = \max(y_{t,l}^p, y_{t,l}^q)$, $x_N = \min(x_{b,r}^p, x_{b,r}^q)$, $y_N = \min(y_{b,r}^p, y_{b,r}^q)$.

The detailed coordinate relation can be referred to in Fig. 6.

2. If $((x_M < x_N) \cap (y_M < y_N))$ is true, the bounding boxes of worker p and window q intersect with each other.

3. Calculate $S_p^i = \frac{(x_M - x_N) \times (y_M - y_N)}{(x_{t,l}^p - x_{b,r}^p) \times (y_{t,l}^p - y_{b,r}^p)}$ if step 2 is satisfied. A part of the computation procedure is shown in Table 3.

The calculated values S_p^i for all the positive samples and negative samples were put into the ASC classifier and the probability density distribution was then analyzed. The density distributions of S_p^i are shown in Fig. 7, from which we get S_r for six categories, respectively. The results are: $S_r^1 = 0.91$, $S_r^2 = 0.90$, $S_r^3 = 0.88$, $S_r^4 = 0.85$, $S_r^5 = 0.83$, $S_r^6 = 0.80$.

4.2. Performance test

To verify the applicability of our method in different scenarios of practical use, we proposed 96 different scenarios (4 illumination conditions $\times$ 4 workers $\times$ 6 windows) to collect diversified testing video clips as shown in Table 4. In each scenario we collected 8–15 video clips to test the detection precision of this scenario. For example, scenario 1

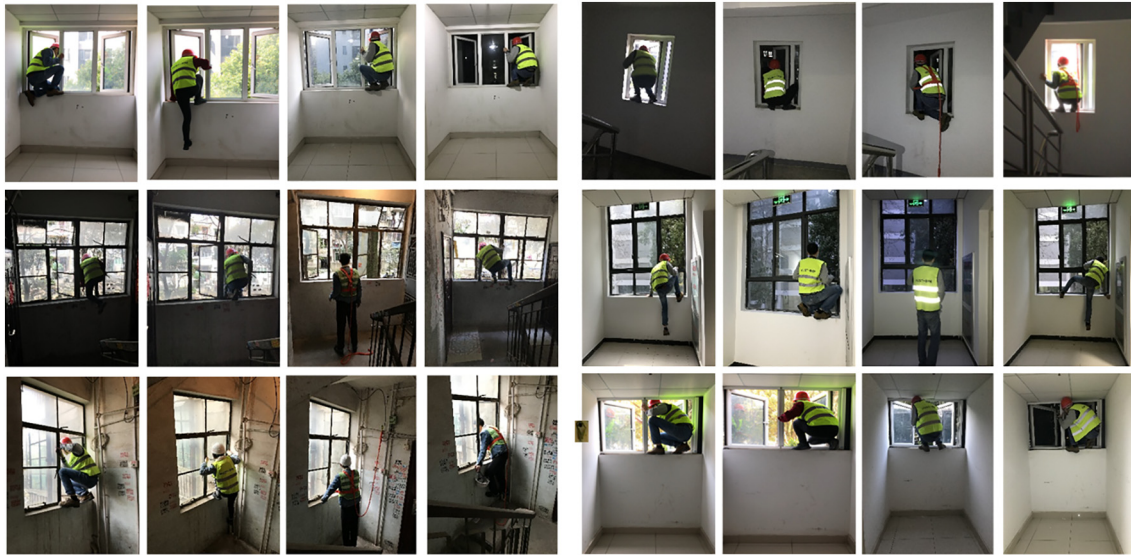


Fig. 12. Image frame examples including different window characteristics.

contains the set of multiple video clips in which worker p_1 is trying to get through a window w_1 in the daylight: 750–1500 (lux) condition. And the video clips in each scenario are different in two aspects: (1) different PPEs use by the worker, (2) a different occlusion ratio caused by the disruption of other workers. These 96 scenarios comprise the testing data for experiment of the impact of varying illumination, physical characteristics of workers and windows. Test results of these 96 scenarios are presented in Table 5. In addition, we collected another 200 video clips where there were severe obstructions, to combine with the 200 video clips that were randomly selected from the above testing data. These 400 video clips comprise the testing data for an experiment of the impact of varying occlusion ratios. Finally, a total of 1316 video clips were collected in the test dataset.

The following paragraphs are the statistical analysis of the testing results. The 96 scenarios (1116 video clips) were divided into different categories according to different testing objectives. For example, to test the impact of different illumination, the 96 scenarios were divided into 4 categories. Such means of classification will help to show the impact of different influences more explicitly. As for the occlusion experiment, we set a special experiment, and the experiment details are shown below.

4.2.1. Impact of illumination

Considering the time of a day, the weather condition and the lighting design of the room greatly influence the illumination in a room, we collected a lot of video clips under different illumination conditions. We used a luminometer to measure the luminous flux of the working environment, and the illumination condition is divided into 4 classes, i.e., 750–1500 (lux) in daylight (bright), 200–750 (lux) in daylight (middle), 20–200 (lux) in daylight (dim), 50–500 (lux) in lamplight (at night), as shown in Fig. 8. The test results show that the illumination hardly affected the video detection. As shown in Fig. 9 and Table 6, the precision and recall rate are almost the same in the visual condition of “200–1500 (lux) in daylight” and “50–500 (lux) in lamplight”, while the performance in low lux condition (under 200 (lux)) in daylight is a bit lower.

4.2.2. Impact of workers' physical characteristics

Diversified appearance features (i.e., height, weight and clothes) and the various postures of workers climbing through the window, might influence the performance of the SSD detector and SORT tracker. An experiment was set to test the existence of this effect. Four workers (shown in Fig. 10) of different appearance and dress were employed to

repeatedly climb through the window and video clips aimed at these scenarios were collected. Results in Fig. 11 and Table 7 show that the individual characteristics had little effect on the performance of our proposed method.

4.2.3. Impact of window characteristics

Different sizes and ratios of windows might cause different positions (see Table. 1) of a steeplejack when climbing through the window, which would lead to different S_r (see Fig. 7). However, whether all of these will finally impact the performance is still unknown and the third experiment was set for this validation. We found six different windows (shown in Fig. 12) to conduct this experiment and workers were asked to repeatedly climb through them. The results in Fig. 13 and Table 8 demonstrate that the impact of window characteristics on performance is negligible.

4.2.4. Impact of occlusion

Construction sites are always occupied by many workers as well as much equipment and materials. The messiness on site might at times cause severe occlusion to the cameras. Since our proposed method is based on surveillance video by cameras, putting equipment and materials neatly is an effective way to avoid occlusion. However, the moving workers inevitably still sometimes cause occlusion (shown in Fig. 14). This paper selects 400 video clips of varying ratios of occlusion to test the impact of occlusion. First, we need to verify the occlusion ratio of each video clip. For every image frame in a video clip, we manually annotated the occlusion ratios of the five target objects (i.e., window, worker, hardhat, harness and anchorage) according to self-subjective judgment, and the ratios of the five were averaged to represent the occlusion ratio of this frame. The occlusion ratio of a video clip is defined as the average value of the occlusion ratios of all the key frames in it. Then, the selected 400 video clips were used to test our method, and the results of all the clips in the same interval of occlusion ratio are computed separately. We defined 10 intervals of occlusion ratio (0–10%, 10%–20%, ..., 90%–100%) and calculated the average value of each interval (5%, 15%, ...) to represent this interval, which is shown in Fig. 15 and Table 9. It's observed that both the precision and recall rate slightly decline as the occlusion ratio increases when the occlusion ratio of a video clip is < 60%. However, when the occlusion ratio exceeds 60%, we observe a dramatic drop in the performance of our method.

Precision [68] is defined as the ratio of TP to TP + FP and measures the reliability of the detection. TP is the number of inappropriate-PPEs-use steeplejacks and where the test results are correct. TP + FP is the



Fig. 13. (a). Precision ratios under different window characteristics
(b). Recall rate ratios under different window characteristics.

Table 8
Precision and recall rate ratios under different window characteristics.

No.	Window	Average precision	Average recall rate
1	W1	0.917	0.942
2	W2	0.910	0.930
3	W3	0.935	0.923
4	W4	0.903	0.943
5	W5	0.933	0.970
6	W6	0.926	0.926

number of steeplejacks detected as inappropriate-PPEs-use based on the method. Recall [68] is the ratio of TP to TP + FN. TP + FN means the actual number of inappropriate-PPEs-use steeplejacks. If one of the PPEs is totally obstructed in a video clip (that is $(\sum E_{hat+}^i = \sum E_{hat-}^i = 0)$ or $(\sum E_{harn+}^i = \sum E_{harn-}^i = 0)$ or $(\sum E_{ach+}^i = \sum E_{ach-}^i = 0)$), there will be no results in the detection. This situation can lead to a missed detection problem for the inappropriate-PPEs-use steeplejack and has a significant impact on the recall rate. On the other hand, the precision

doesn't change as much as the recall rate since occlusion won't cause misjudgments with our method. However, considering that occlusion will reduce the number of key video frames, it might cause insufficient samples to judge the difference between $\sum E_+$ and $\sum E_-$ and finally lead to a slight decrease in precision.

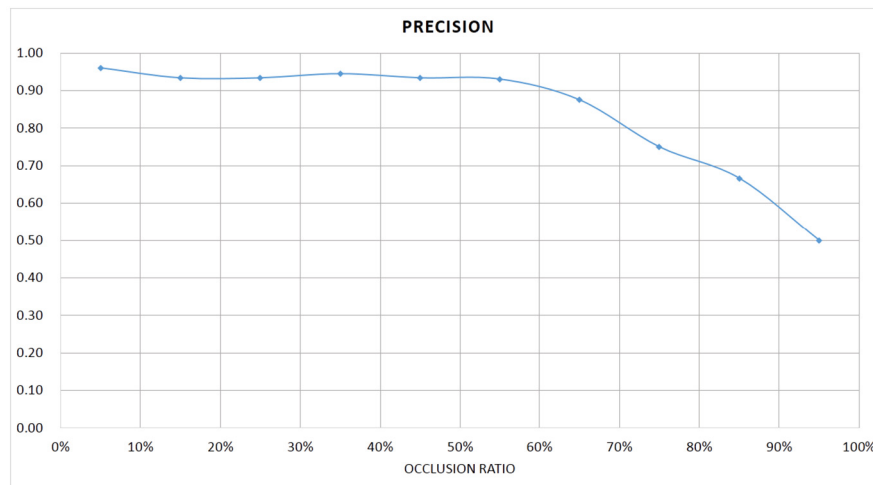
5. Discussion

First, it noted that this paper is aimed at regular windows instead of sliding glass doors, since the aerial scenarios recognition method is based on spatial interactions between a worker and a window, which doesn't apply to sliding glass doors. Moreover, outside the sliding glass door is a balcony instead of a suspended environment, which is different from our hypothesis. Furthermore, our method generally applies to high-rise buildings with many regular windows, such as the Public Rental Housing and Home Ownership Scheme Housing in Hong Kong.

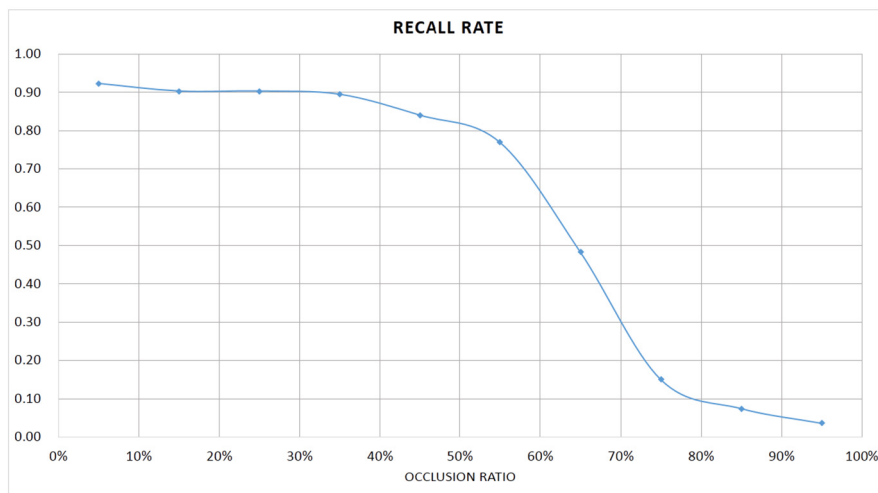
In this section we will verify the feasibility of the proposed method in terms of technical applicability and cost. Multiple possible solutions (indoor positioning techniques and manual supervision) are presented



Fig. 14. Image frame examples including different occlusion rate.



(a).



(b).

Fig. 15. (a). Precision ratios under different occlusion rate
(b). Recall rate ratios under different occlusion rate.

and compared to our method in the following paragraphs.

Indoor positioning is a system to locate the workers and objects using radio waves or other sensory information. Since the purpose of this study is to monitor the PPEs use of steeplejacks, the positioning techniques are required to first determine whether the steeplejack is indoors or outside the building (in an aerial working area). Considering that the thickness of the wall is only about 0.2 m, the adopted positioning technique won't be applicable in this study unless its accuracy achieves < 0.2 m. Although the newest technologies of indoor positioning can achieve enough precision for distinguishing between outdoors and indoors in dense passive RFID tags environment [69–72], the

application of these techniques in our study are facing several challenges. First, it would be a large expense and heavy work to build and maintain such a dense RFID tags environment. It's noted that the changing working environment in the room greatly influence the positioning precision and requires frequent calibration. Second, the dense deployment of RFID tags may influence daily construction work in room. The dense tags are also easy to be damaged by the frequent movements of workers.

Compared to indoor positioning, our proposed method is able to accurately recognize those who are ready for outside work with lower cost, less maintenance effort and less disruption to the construction

Table 9
Test results under different occlusion rates.

No.	Occlusion rate	Number of video clips	Number of inappropriate-PPEs-use steeplejacks	TP	FP	FN	Precision	Recall rate
1	5%	28	26	24	1	2	0.960	0.923
2	15%	34	31	28	2	3	0.933	0.903
3	25%	37	31	28	2	3	0.933	0.903
4	35%	69	57	51	3	6	0.944	0.895
5	45%	55	50	42	3	8	0.933	0.840
6	55%	59	52	40	3	12	0.930	0.769
7	65%	29	29	14	2	15	0.875	0.483
8	75%	24	20	3	1	17	0.750	0.150
9	85%	33	27	2	1	25	0.667	0.074
10	95%	32	28	1	1	27	0.500	0.036

work, although several limitations still exist in its applicability.

First, the performance of our method is highly robust to the change of illumination, individual characteristics and window characteristics. The results show that they all achieve an average precision and recall of over 0.9 in the above experiments. As to occlusion problems, since we have combined the results of multiple frames to determine the behavior of steeplejacks, the impact of occlusion has been greatly reduced. As long as at least one E_{hat+}^i or E_{hat-}^i is detected in the video clip, we can almost determine if the steeplejacks have been wearing a hardhat, and similarly for the inspection of harness and anchorage usage. However, the performance of our method can be slightly improved if the occlusion ratio of a video clip decreases (see Fig. 15).

The second limitation of our method is that it needs to put at least one camera per room for automated surveillance. It seems difficult, but it's still realistic in practical use. Considering that a room won't be as wide as the construction site, an ordinary camera with lower resolution is enough to meet the requirements for monitoring. The price of this kind of camera is about 200 HKD. Suppose there are 10 rooms that need to be monitored, the total cost of the cameras will only be about 2000 HKD. Cameras are one-time investments and all the cameras can be recycled or preserved for later use. However, if we employ a surveyor to manually monitor the behavior of steeplejacks in 10 rooms, the labor cost will be at least 20,000 HKD per month according to the minimum wage level for the construction sites of Hong Kong. It should, though, be noted that the surveyor would find it very hard to manage since he would have to monitor 10 rooms at the same time.

Third, the inspection for anchorage use by our method might not always be right. If the anchorage is linked with another webbing that doesn't belong to a steeplejack, this steeplejack is still mistaken for having hung his webbing correctly. This situation is the result of the separate judgment of two conditions; whether a steeplejack is wearing a harness and whether an anchorage is linked with a webbing. Another solution is to view the harness, webbing and anchorage as a whole. A detection algorithm is employed to analyze this ensemble. However, a bounding box for this ensemble is an oversized area with a high proportion of useless background, which brings a lot of noise and disruption to the extracted features and is not an ideal solution to ensure the detection of the ensemble. Viewed from another perspective, few workers would hang their safety belts on the anchorage and leave.

This study has made three major contributions to knowledge. Firstly, we proposed a method to prevent falling accidents from a new perspective of checking the appropriate usage of PPEs for the steeplejacks, which is different from other researchers' solutions. We have considered a thorough inspection program by detecting six different scenarios for steeplejacks, which not only requires the wearing of a harness, but also to hang the webbing on the anchorage and wear hardhats. Secondly, we proposed an aerial operation scenario understanding method by combining the computer vision technologies and an ASC classifier to detect steeplejacks. The scenario understanding method makes the inspection a preventative control measure and more applicable to the site. Thirdly, the PPEs checking method can effectively

mitigate occlusion influence. Also the inspection of PPEs includes not only harness checking but also webbing, anchoring and wearing hardhats. Moreover, experiment results indicate that our framework eliminates the impact of the noise and error that was generated in vision based detection and tracking algorithms, and achieves a favorable performance.

6. Conclusion

Despite various research being devoted to fall prevention, such as the removal of hazardous areas and the use of guardrails or safety net systems, little research focuses on supervising the appropriate use of PPEs. PPEs are a last protection measure that guarantees the safety of steeplejacks in case accidents happen, but they don't attract enough attention from researchers. This paper proposed an automated inspection method based on deep-learning assisted computer vision technology for the safety measures of steeplejacks. We considered that the best solution to prevent fall accidents is to thoroughly inspect the safety measures of steeplejacks before they enter an aerial working area. Therefore, we proposed an ASC classifier to detect the steeplejacks who are going through a window into an aerial working area. Then, an advanced object detector SSD and CNN classifier were employed in this paper to judge whether a steeplejack has been made adequately safe.

Due to advanced and thorough learning methods and the high resolution of the images, we achieved a high precision and recall rate. Also, the experiment results show that our method is robust to most changing scenarios, such as illumination, individual and window characteristics, and partial occlusion (our method still achieves high performance when the occlusion ratio is < 0.6). We make a detailed comparison with other solutions in terms of technical applicability and cost, including indoor positioning and manual checking. Also, a computer vision-based solution proves to be the most cost-effective solution for solving safety monitoring of aerial work. Another limitation of this paper is that it can't distinguish between the anchorages that link to empty webbings from the ones that connect to the workers' harnesses. Considering that this would mean that the workers were cheating the method, and such behavior would not benefit them but put them in danger, it's expected that workers would rarely take such actions. We recommend that future research focuses on reward and punishment of steeplejacks after the detection of inappropriate use of PPEs using our method.

Acknowledgement

We are thankful for the financial support of 1) the National 12th Five-Year Plan Major Scientific and Technological Issues (NFYPMSTI) through Grant 2015BAK33B04; 2) The Research Grants Council of Hong Kong grant entitled "Proactively Monitoring Construction Progress by Integrating 3D Laser-scanning and BIM" (PolyU 152093/14E); and 3) National Science Foundation of China (grant no.51678265).

References

- [1] Occupational Safety and Health Administration, OSHA's Fall Prevention Campaign, <https://www.osha.gov/stopfalls/index.html>, (2017) (Last accessed on 8 September 2017).
- [2] Bureau of Labor Statistics, Census of Fatal Occupational Injuries Summary, 2015, <https://www.bls.gov/news.release/foi.nr0.htm>, (2016) (Last accessed on 8 September 2017).
- [3] Health and Safety Executive, Kinds of Accident in Great Britain, 2016, <http://www.hse.gov.uk/statistics/causinj/kinds-of-accident.pdf>, (2016) (Last accessed on 8th September 2017).
- [4] Health and Safety Executive Fatal Injuries Arising From Accidents at Work in Great Britain: Headline Results, <http://www.hse.gov.uk/statistics/fatals.htm>, (2017) (Last accessed on 13 Nov 2017).
- [5] National Institute for Occupational Safety and Health, Research and Practice for Fall Injury Control in the Workplace: Proceedings of International Conference on Fall Prevention and Protection, <https://www.cdc.gov/niosh/docs/2012-103/pdfs/2012-103.pdf>, (2010) (Last accessed on 13 Nov 2017).
- [6] Commonly Used Statistics, Occupational Safety and Health Administration, (2017) <https://www.osha.gov/oshstats/commonstats.html> (Last accessed on 13 Nov 2017).
- [7] Hong Kong Housing Authority, Practical Guide to Working at Height: Ensuring Safe Work Practices, http://www.housingauthority.gov.hk/mini-site/site-safety/common/resources/article/pdf/publications/safety-handbooks-and-booklets/Eng_web_version.pdf, (2012) (Last accessed on 8th September 2017).
- [8] Occupational Safety and Health Administration, Duty to Have Fall Protection, https://www.osha.gov/pls/oshaweb/owasrch.search_form?p_doc_type=INTERPRETATIONS&p_toc_level=3&p_keyvalue=1926.501&p_status=CURRENT (Last accessed on 29 March 2018).
- [9] Occupational Safety and Health Administration, Fall Protection Systems Criteria and Practices, https://www.osha.gov/pls/oshaweb/owasrch.search_form?p_doc_type=INTERPRETATIONS&p_toc_level=3&p_keyvalue=1926.502&p_status=CURRENT (Last accessed on 29 March 2018).
- [10] The Work at Height Regulations, <http://www.legislation.gov.uk/uk/si/2005/735/introduction/made>, (2005) (Last accessed on 29 March 2018).
- [11] Personal Fall Arrest Systems, <https://www.osha.gov/SLTC/etools/construction/falls/fallarrest.html> (Last accessed on 29 March 2018).
- [12] X.S. Dong, J.A. Largay, S.D. Choi, X. Wang, C.T. Cain, N. Romano, Fatal falls and PFAS use in the construction industry: findings from the NIOSH FACE reports, *Accid. Anal. Prev.* 102 (2017) 136–143, <http://dx.doi.org/10.1016/j.aap.2017.02.028>.
- [13] Occupational Safety & Health Administration, Determining the Need for Hard Hat and Eye Protection on Construction Sites, https://www.osha.gov/pls/oshaweb/owasrch.search_form?p_doc_type=INTERPRETATIONS&p_toc_level=3&p_keyvalue=1926.100&p_status=CURRENT, (2004) (Last accessed on 29 March 2018).
- [14] Health and Safety Executive, Causal Factors in Construction Accidents, <http://www.hse.gov.uk/research/rrpdf/rr156.pdf>, (2003) (Last accessed on 8th September 2017).
- [15] R. K., J. N., Occupational accident patterns and prevention measures in construction sites in Nairobi County Kenya, *Am. J. Civ. Eng.* 4 (5) (2016) 254–263.
- [16] Bureau of Labor Statistics, National Census of Fatal Occupational Injuries in 2011, https://www.bls.gov/news.release/archives/foi_09202012.pdf, (2011) (Last accessed on 8 September 2017).
- [17] Construction eTool-Falls, <https://www.osha.gov/SLTC/etools/construction/falls/mainpage.html> (Last accessed on 14 September 2017).
- [18] B. Esmail, M. Halliwell, M. Rouchery, Developing a framework for measuring the effectiveness of common fall prevention/protection practices, *International Conference on Sustainable Design, Engineering, and Construction 2012*, 2012, pp. 719–726, <http://dx.doi.org/10.1061/9780784412688.086>.
- [19] J. Qi, R.R. Issa, S. Olbina, J. Hinze, Use of building information modeling in design to prevent construction worker falls, *J. Comput. Civ. Eng.* 28 (5) (2013) A4014008, [http://dx.doi.org/10.1061/\(ASCE\)CP.1943-5487.0000365](http://dx.doi.org/10.1061/(ASCE)CP.1943-5487.0000365).
- [20] S. Zhang, K. Sulankivi, M. Kiviniemi, I. Romo, C.M. Eastman, J. Teizer, BIM-based fall hazard identification and prevention in construction safety planning, *Saf. Sci.* 72 (2015) 31–45, <http://dx.doi.org/10.1016/j.ssci.2014.08.001>.
- [21] J. Wang, N. Pradhananga, J. Teizer, Automatic fall risk identification using point cloud data in construction excavation, *Computing in Civil and Building Engineering*, 2014, pp. 981–988, <http://dx.doi.org/10.1061/9780784413616.122>.
- [22] R. Navon, O. Kolton, Algorithms for automated monitoring and control of fall hazards, *J. Comput. Civ. Eng.* 21 (1) (2007) 21–28, [http://dx.doi.org/10.1061/\(ASCE\)0887-3801\(2007\)21:1\(21\)](http://dx.doi.org/10.1061/(ASCE)0887-3801(2007)21:1(21)).
- [23] R. Navon, O. Kolton, Model for automated monitoring of fall hazards in building construction, *J. Constr. Eng. Manag.* 132 (7) (2006) 733–740, [http://dx.doi.org/10.1061/\(ASCE\)0733-9364\(2006\)132:7\(733\)](http://dx.doi.org/10.1061/(ASCE)0733-9364(2006)132:7(733)).
- [24] C.M. Zuluaga, A. Albert, Preventing falls: choosing compatible fall protection supplementary devices (FPSD) for bridge maintenance work using virtual prototyping, *Saf. Sci.* (2017), <http://dx.doi.org/10.1016/j.ssci.2017.08.006>.
- [25] E. Cheung, A.P. Chan, Rapid demountable platform (RDP)—a device for preventing fall from height accidents, *Accid. Anal. Prev.* 48 (2012) 235–245, <http://dx.doi.org/10.1016/j.aap.2011.05.037>.
- [26] V. Kaskutas, A.M. Dale, H. Lipscomb, B. Evanoff, Fall prevention and safety communication training for foremen: report of a pilot project designed to improve residential construction safety, *J. Saf. Res.* 44 (2013) 111–118, <http://dx.doi.org/10.1016/j.jsr.2012.08.020>.
- [27] K. Lin, G. Migliaccio, R. Azari, C. Lee, J. De la Lata, Developing 3D safety training materials on fall related hazards for limited English proficiency (LEP) and low literacy (LL) construction workers, *J. Comput. Civ. Eng.* (2012) 113–120, <http://dx.doi.org/10.1061/9780784412343.0015>.
- [28] B. Evanoff, A.M. Dale, A. Zeringue, M. Fuchs, J. Gaal, H.J. Lipscomb, V. Kaskutas, Results of a fall prevention educational intervention for residential construction, *Saf. Sci.* 89 (2016) 301–307, <http://dx.doi.org/10.1016/j.ssci.2016.06.019>.
- [29] A.L. Cheng, C. Georgoulas, T. Bock, Fall detection and intervention based on wireless sensor network technologies, *Autom. Constr.* 71 (2016) 116–136, <http://dx.doi.org/10.1016/j.autcon.2016.03.004>.
- [30] H. Jebelli, C.R. Ahn, T.L. Stentz, Comprehensive fall-risk assessment of construction workers using inertial measurement units: validation of the gait-stability metric to assess the fall risk of iron workers, *J. Comput. Civ. Eng.* 30 (3) (2015) 04015034, [http://dx.doi.org/10.1061/\(ASCE\)CP.1943-5487.0000511](http://dx.doi.org/10.1061/(ASCE)CP.1943-5487.0000511).
- [31] K. Yang, C.R. Ahn, M.C. Vuran, S.S. Aria, Semi-supervised near-miss fall detection for ironworkers with a wearable inertial measurement unit, *Autom. Constr.* 68 (2016) 194–202, <http://dx.doi.org/10.1016/j.autcon.2016.04.007>.
- [32] K. Yang, S. Aria, C.R. Ahn, T.L. Stentz, Automated detection of near-miss fall incidents in iron workers using inertial measurement units, *Construction Research Congress 2014*, 2014, pp. 935–944, <http://dx.doi.org/10.1061/9780784413517.096>.
- [33] K. Yang, H. Jebelli, C. Ahn, M. Vuran, Threshold-based approach to detect near-miss falls of iron workers using inertial measurement units, *International Workshop on Computing in Civil Engineering*, 2015, pp. 148–155, <http://dx.doi.org/10.1061/9780784479247.019>.
- [34] Personal protective equipment, https://en.wikipedia.org/wiki/Personal_protective_equipment (Last accessed on 29 March 2018).
- [35] S. Barro-Torres, T.M. Fernández-Caramés, H.J. Pérez-Iglesias, C.J. Escudero, Real-time personal protective equipment monitoring system, *Comput. Commun.* 36 (1) (2012) 42–50, <http://dx.doi.org/10.1016/j.comcom.2012.01.005>.
- [36] A. Kelm, L. Laußat, A. Meins-Becker, D. Platz, M.J. Khazaei, A.M. Costin, M. Helmus, J. Teizer, Mobile passive radio frequency identification (RFID) portal for automated and rapid control of personal protective equipment (PPE) on construction sites, *Autom. Constr.* 36 (2013) 38–52, <http://dx.doi.org/10.1016/j.autcon.2013.08.009>.
- [37] D. Podgórski, K. Majchrzycka, A. Dąbrowska, G. Gralewicz, M. Okrasa, Towards a conceptual framework of OSH risk management in smart working environments based on smart PPE, ambient intelligence and the internet of things technologies, *Int. J. Occup. Saf. Ergon.* 23 (1) (2017) 1–20, <http://dx.doi.org/10.1080/10803548.2016.1214431>.
- [38] S. Dong, Q. He, H. Li, Q. Yin, Automated PPE misuse identification and assessment for safety performance enhancement, *International Conference on Construction and Real Estate Management*, 2015, pp. 204–214, <http://dx.doi.org/10.1061/9780784479377.024>.
- [39] S. Du, M. Shehata, W. Badawy, Hard hat detection in video sequences based on face features, motion and color information, *3rd International Conference on Computer Research and Development*, Vol. 4 IEEE, 2011, pp. 25–29, <http://dx.doi.org/10.1109/ICCRD.2011.5763846>.
- [40] K. Shrestha, P.P. Shrestha, D. Bajracharya, E.A. Yfantis, Hard-hat detection for construction safety visualization, *J. Constr. Eng. Manag.* 2015 (2015), <http://dx.doi.org/10.1155/2015/721380>.
- [41] M.-W. Park, N. Elsafty, Z. Zhu, Hardhat-wearing detection for enhancing on-site safety of construction workers, *J. Constr. Eng. Manag.* 141 (9) (2015) 04015024, [http://dx.doi.org/10.1061/\(ASCE\)CO.1943-7862.0000974](http://dx.doi.org/10.1061/(ASCE)CO.1943-7862.0000974).
- [42] P. Ott, M. Everingham, Implicit color segmentation features for pedestrian and object detection, *IEEE 12th International Conference on Computer Vision*, IEEE, 2009, pp. 723–730, <http://dx.doi.org/10.1109/ICCV.2009.5459238>.
- [43] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, *Autom. Constr.* 85 (2018) 1–9, <http://dx.doi.org/10.1016/j.autcon.2017.09.018>.
- [44] M.J. Skibniewski, Research trends in information technology applications in construction safety engineering and management, *Front. Eng. Manag.* 1 (3) (2014) 246–259, <http://dx.doi.org/10.15302/j-fem-2014034>.
- [45] H. Guo, Y. Yu, M. Skitmore, Visualization technology-based construction safety management: a review, *Autom. Constr.* 73 (2017) 135–144, <http://dx.doi.org/10.1016/j.autcon.2016.10.004>.
- [46] J. Seo, S. Han, S. Lee, H. Kim, Computer vision techniques for construction safety and health monitoring, *Adv. Eng. Inform.* 29 (2) (2015) 239–251, <http://dx.doi.org/10.1016/j.aei.2015.02.001>.
- [47] L. Ding, W. Fang, H. Luo, P.E. Love, B. Zhong, X. Ouyang, A deep hybrid learning model to detect unsafe behavior: integrating convolution neural networks and long short-term memory, *Autom. Constr.* 86 (2018) 118–124.
- [48] Q. Fang, H. Li, X. Luo, L. Ding, T.M. Rose, W. An, Y. Yu, A deep learning-based method for detecting non-certified work on construction sites, *Adv. Eng. Inform.* 35 (2018) 56–68, <http://dx.doi.org/10.1016/j.aei.2018.01.001>.
- [49] W. Fang, L. Ding, B. Zhong, P.E. Love, H. Luo, Automated detection of workers and heavy equipment on construction sites: a convolutional neural network approach, *Adv. Eng. Inform.* 37 (2018) 139–149, <http://dx.doi.org/10.1016/j.aei.2018.05.003>.
- [50] Y. LeCun, Y. Bengio, G. Hinton, Deep learning, *Nature* 521 (7553) (2015) 436–444, <http://dx.doi.org/10.1038/nature14539>.
- [51] R. Girshick, J. Donahue, T. Darrell, J. Malik, Rich feature hierarchies for accurate object detection and semantic segmentation, *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* (2014) 580–587, <http://dx.doi.org/10.1109/CVPR.2014.81>.
- [52] K. He, X. Zhang, S. Ren, J. Sun, Spatial pyramid pooling in deep convolutional

- networks for visual recognition, European Conference on Computer Vision, Springer, 2014, pp. 346–361, , <http://dx.doi.org/10.1109/TPAMI.2015.2389824>.
- [53] S. Ren, K. He, R. Girshick, J. Sun, Faster R-CNN: towards real-time object detection with region proposal networks, *Adv. Neural Inf. Proces. Syst.* (2015) 91–99, <http://dx.doi.org/10.1109/TPAMI.2016.2577031>.
- [54] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: unified, real-time object detection, *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.* (2016) 779–788, <http://dx.doi.org/10.1109/CVPR.2016.91>.
- [55] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, A.C. Berg, SSD: Single shot multibox detector, *European Conference on Computer Vision*, Springer, 2016, pp. 21–37.
- [56] Z. Bolei, A. Khosla, A. Lapedriza, A. Oliva, A. Torralba, Object Detectors Emerge in Deep Scene CNNs, (2015).
- [57] A. Bewley, Z. Ge, L. Ott, F. Ramos, B. Upcroft, Simple online and realtime tracking, *IEEE International Conference on Image Processing*, IEEE, 2016, pp. 3464–3468, , <http://dx.doi.org/10.1109/ICIP.2016.7533003>.
- [58] W. Choi, Near-online multi-target tracking with aggregated local flow descriptor, *IEEE International Conference on Computer Vision*, 2015, pp. 3029–3037, , <http://dx.doi.org/10.1109/ICCV.2015.347>.
- [59] C. Kim, F. Li, A. Ciptadi, J.M. Rehg, Multiple hypothesis tracking revisited, *IEEE International Conference on Computer Vision*, 2015, pp. 4696–4704, , <http://dx.doi.org/10.1109/ICCV.2015.533>.
- [60] L. Leal-Taixé, A. Milan, I. Reid, S. Roth, K. Schindler, MottChallenge 2015: towards a benchmark for multi-target tracking, *IEEE Conference on Computer Vision and Pattern Recognition*, 2015.
- [61] J.H. Yoon, M.-H. Yang, J. Lim, K.-J. Yoon, Bayesian multi-object tracking using motion context from multiple objects, *IEEE Winter Conference on Applications of Computer Vision*, IEEE, 2015, pp. 33–40, , <http://dx.doi.org/10.1109/WACV.2015.12>.
- [62] R.E. Kalman, A new approach to linear filtering and prediction problems, *J. Basic Eng.* 82 (1) (1960) 35–45, <http://dx.doi.org/10.1115/1.3662552>.
- [63] H.W. Kuhn, The Hungarian method for the assignment problem, *Nav. Res. Logist.* 2 (1–2) (1955) 83–97, <http://dx.doi.org/10.1002/nav.3800020109>.
- [64] S. Aarshi, S. Chitrakala, Scene Understanding—A Survey, 2017 International Conference on Computer, Communication and Signal Processing, IEEE, 2017, pp. 1–4, <http://dx.doi.org/10.1109/ICCCSP.2017.7944094>.
- [65] G. Papadopoulos, P.J. Edwards, A.F. Murray, Confidence estimation methods for neural networks: a practical comparison, *IEEE Trans. Neural Netw.* 12 (6) (2001) 1278–1287, <http://dx.doi.org/10.1109/72.963764>.
- [66] J.D. Rennie, L. Shih, J. Teevan, D.R. Karger, Tackling the poor assumptions of naive bayes text classifiers, *Proceedings of the 20th International Conference on Machine Learning (ICML-03)*, 2003, pp. 616–623.
- [67] Origin, <https://www.originlab.com/>, (2018) (Last accessed on 29 March 2018).
- [68] D.M. Powers, Evaluation: from precision, recall and F-measure to ROC, informedness, markedness and correlation, *J. Mach. Learn. Technol.* 2 (2011) 37–63.
- [69] Z. Wang, N. Ye, R. Malekian, F. Xiao, R. Wang, TrackT: accurate tracking of RFID tags with mm-level accuracy using first-order Taylor series approximation, *Ad Hoc Netw.* 53 (2016) 132–144, <http://dx.doi.org/10.1016/j.adhoc.2016.09.026>.
- [70] Research and Practice for Fall Injury Control in the Workplace, <https://www.cdc.gov/niosh/docs/2012-103/pdfs/2012-103.pdf> (Last accessed on 29 March 2018).
- [71] D. Zhang, L.T. Yang, M. Chen, S. Zhao, M. Guo, Y. Zhang, Real-time locating systems using active RFID for internet of things, *IEEE Syst. J.* 10 (3) (2016) 1226–1235, <http://dx.doi.org/10.1109/JSYST.2014.2346625>.
- [72] T. Liu, L. Yang, Q. Lin, Y. Guo, Y. Liu, Anchor-free backscatter positioning for RFID tags with high accuracy, *IEEE Conference on Computer Communications*, 2014, pp. 379–387, , <http://dx.doi.org/10.1109/INFOCOM.2014.6847960>.